5) Naive Bayes ⇒

29/01/2026

Naive Bayes ⇒ ALGORITHM

True ✓

Agenda →

- → Naive Bayes (Ps)
- → understanding
- → probabilities
- → Time / Space complexity
- → implementation

Bayes theorem

Naive Bayes

↳ simplest

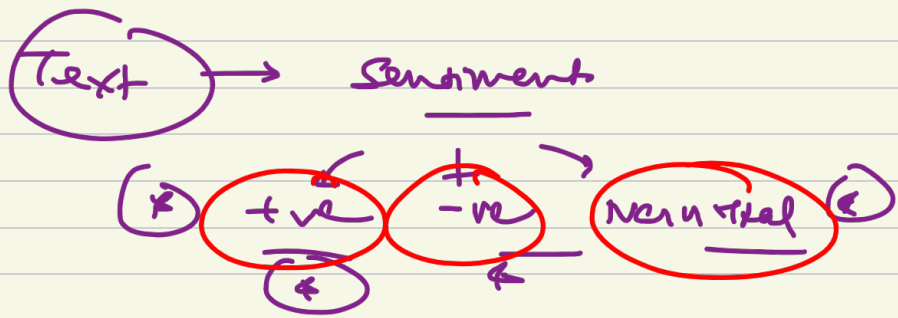
Textual data → works fairly well

Baseline metric

Linear Reg → mean model

(*) passing marks

1st algo → Sentiment Analysis

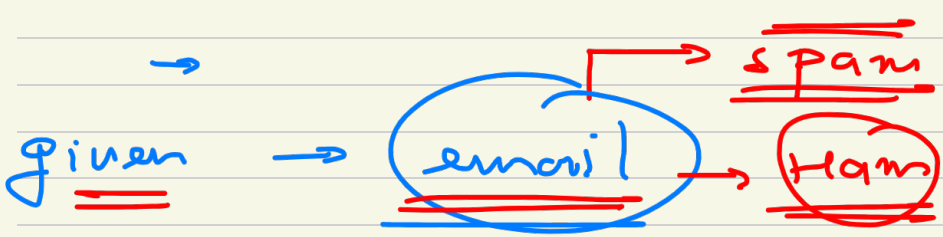


Problem statement =>

(*) (Spam Classification)

gmail → Spam detection

→



1. keywords ⇒ $x_1, x_2, \ldots$ ⇒

2. promotions, address

3. domain, images, free

E_1 → I nigerian prince need your help. send money

E_2 → Meeting scheduled at 5 pm

E_1 ⇒ spam

(lottery, click, money, win, free)

Email → spam

⑧ Text Classification

Email → Spam | Ham

① Sentiment analysis → Text Class

Reviews of any product

Product → good or bad

→ Supervised

Email	class
1 — Text	1
2 — Text	0
⋮	⋮
n	

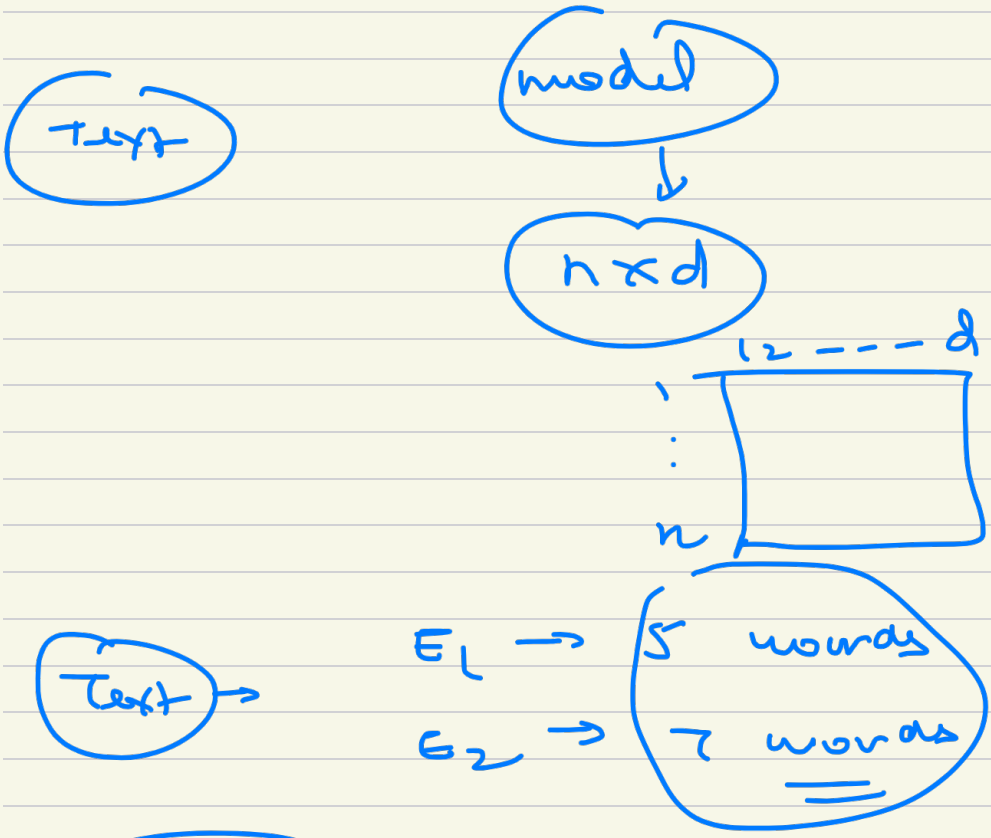
text = < w_1 w_2 ... w_d >

↓ input ⑧

→ Biggest problem of NLP?

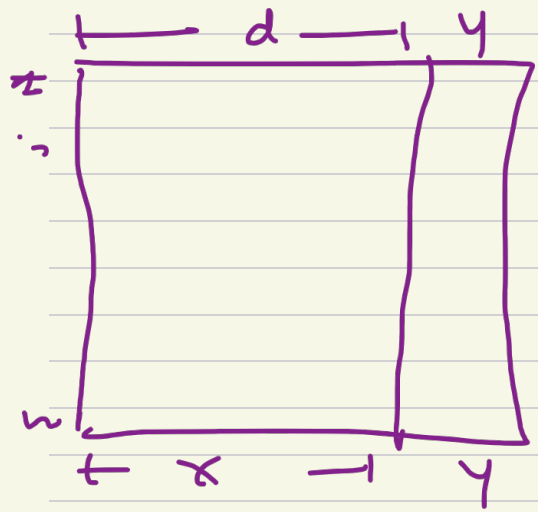
2 emails → X same length

X



7 Textual Pre processing

1. Tokenization
2. Lowercase
3. Punctuations
4. Stopwords
5. VECTORS

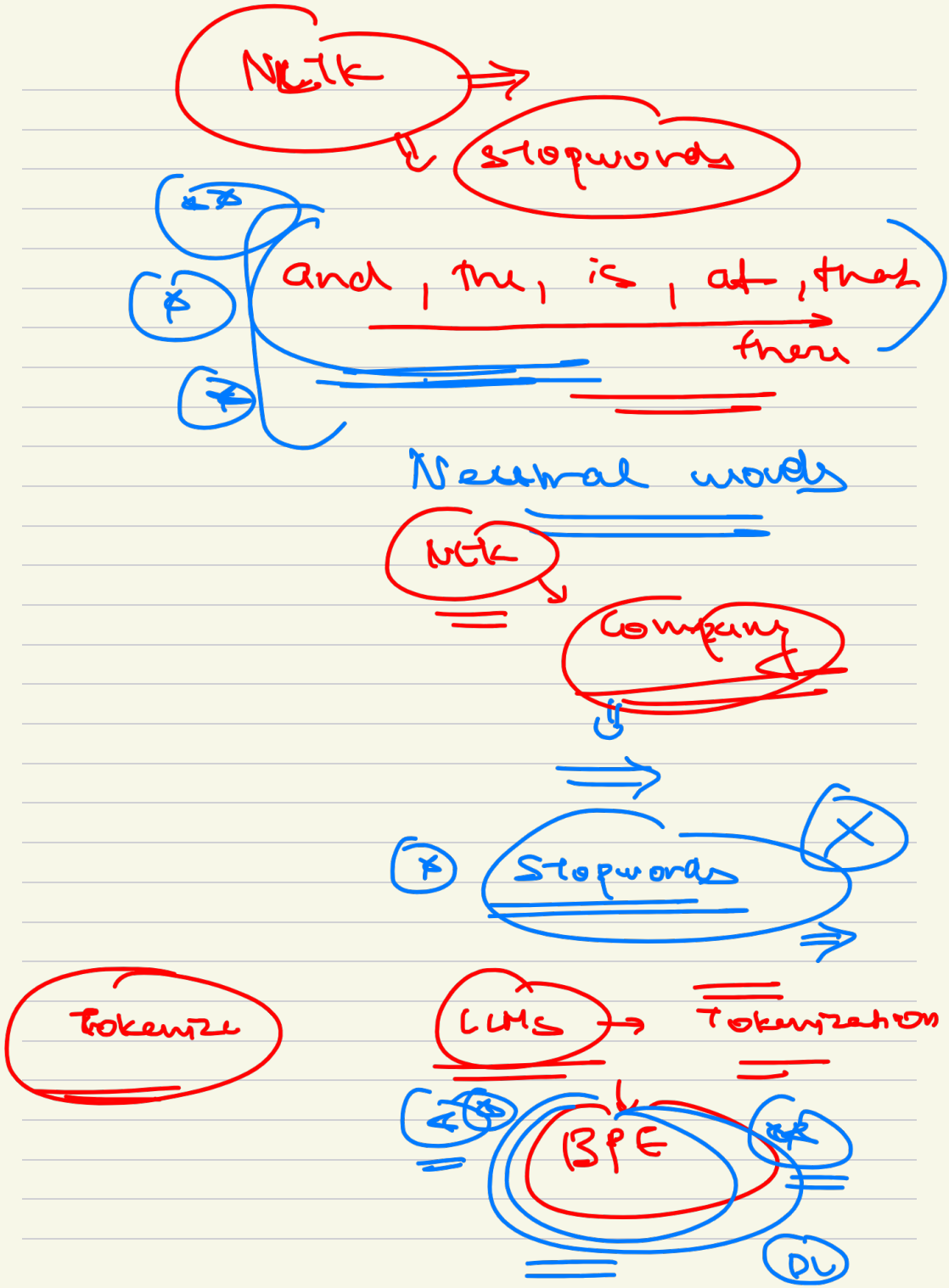


X d

ML

Structured

E1 → Meeting scheduled at 5 pm


NLTK
Stopwords
and , the , is , at , that , there
Normal words
NLTK
Company
Stopwords
Tokenize
LLMs
Tokenization
BPE
DL

Naive Bayes intuition

text i

y = 1 → Spam

y = 0 → Not spam
Ham

P(y = 1 | text i) →

P(y = 0 | text i) →

reg → text

text →

P(y = 1 | text i) → 82%

P(y = 0 | text i) → 18%

(2)

Bayes Theorem

$$P(y=1 \mid w_1, w_2, w_3 \dots w_d)$$

$$P(A \mid B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

- $P(B/A)$ → likelihood
- $P(A)$ → prior
- $P(B)$ → Evidence
- $P(A \mid B)$ → Posterior prob.

Bayes ⇒ Prob.

$P(y=1 \mid \text{text}_i)$

$P(y=0 \mid \text{text})$

$P(y=1 \mid w_1\ w_2 \cdots w_d)$ — A | B

$= P(w_1\ w_2 \cdots w_d \mid y=1) \cdot P(y=1)$ — B | A

$P(w_1\ w_2 \cdots w_d)$

Class 0 → Not spam

Ham

$P(y=0 \mid w_1, w_2 \ldots w_d)$

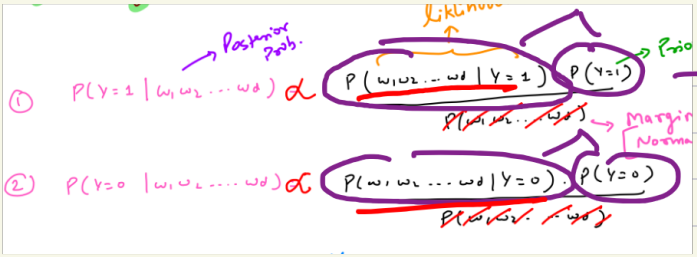
$= P(w_1, w_2 \ldots w_d \mid y=0) \cdot P(y=0)$

$P(w_1, w_2 \ldots w_d)$

Posterior prob.

ignore denom

$\frac{52}{2} > \frac{16}{2}$ ✓



Quiz time!

Time Left: 4s

Let's say:
the probability of dangerous fires are rare (1%)
but smoke is fairly common (10%) due to barbecues,
and 90% of dangerous fires make smoke
Can you find the probability of dangerous Fire when there is Smoke?

22 users have participated

A	0.09	73%
B	0.9	18%
C	0.1	9%

End Quiz Now

$P(A) \Rightarrow P(\text{fire})$

$P(B) = P(\text{smoke})$

$P(S \mid F) = 90\%$.

$P(F \mid S) = ?$

$$P(F \mid S) = \frac{P(S \mid F) \cdot P(F)}{P(S)}$$

0.9 $\quad$ 0.01 $\quad$ 0.1

$= 0.09$

$P(y=1)$, $P(y=0)$

↓

How many

spam out of all

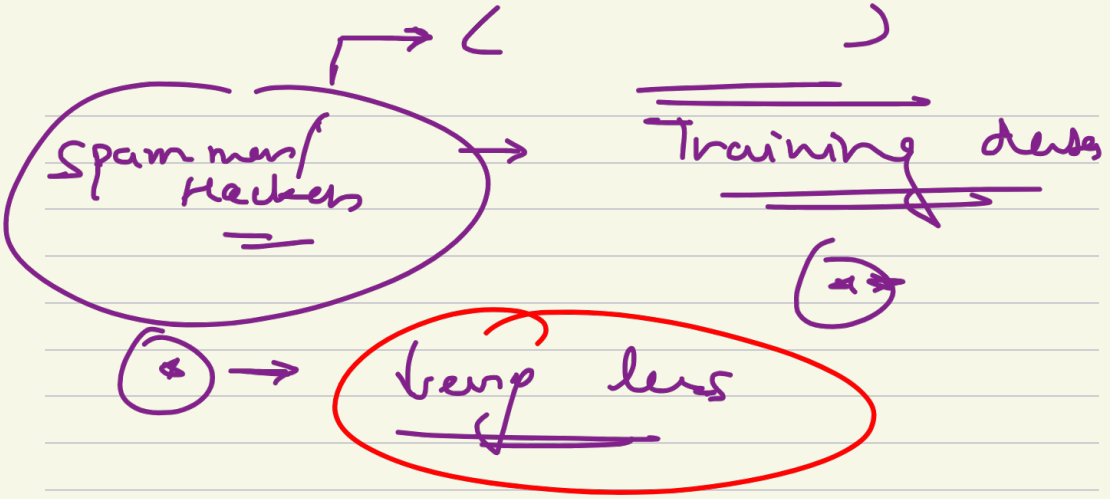
10:1 $

Likelihood → conditional prob.

Naive Bayes Assumption

→ not true for Real world

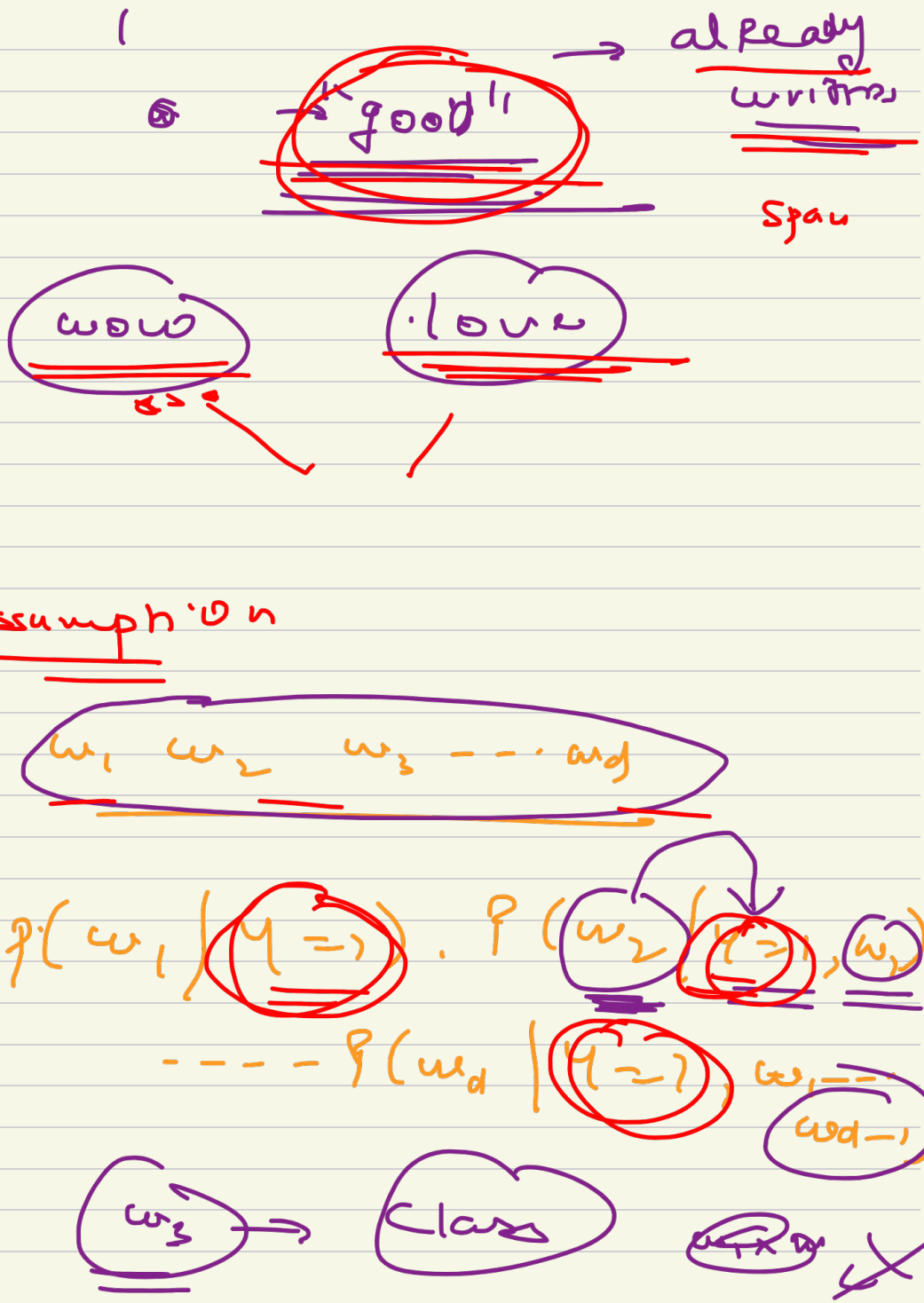
$P(\text{text} \mid y=1) = P(w_1, w_2 \cdots w_d \mid y=1)$ (spam)



Assumption → $w_1, w_2, \ldots w_d$ are independent, condition on class label

$$P(w_1 \mid y=1) \cdot P(w_2 \mid y=1, w_1) \cdots P(w_d \mid y=1, w_1, \ldots w_{d-1})$$

Tedious task


already
written
Spam
assumption

$w_d \Rightarrow$ Class $\times$ $w_1 \cdots w_{d-1}$

word → independent

Assumption — Naive assumption

(*) Simplifying the text ⇒

$P(\text{text} \mid y=1) = P(w_1 \mid y=1) \cdot$

$P(w_2 \mid y=1) \cdot$

$\cdots P(w_d \mid y=1)$

$$P(w_1, w_2 \cdots w_d \mid y=1) = \prod_{i=1}^{d} P(w_i \mid y=1)$$

Spam

word → "free"

$P(\text{word} = \text{free} \mid \text{spam})$

$$= \frac{\#\text{Spam emails with "free"}}{\#\text{Spam emails}}$$

$P(\omega_2 \mid y=0) =$ FREE

(*)

How to finally make the pred^n $\Rightarrow$

$$P(y=1 \mid \text{text}) = \frac{P(\text{text} \mid y=1) \cdot P(y=1)}{P(\text{Text})}$$

$P(\text{Text})$ ignore

$$P(y=1) \cdot \prod_{i=1}^{d} P(w_i \mid y=1)$$

$w_1 \rightarrow$ spam

$w_2 \rightarrow$ spam

$w_d \rightarrow$ spam

① P(Y=0)

P(Y=1 | text)

↓

P(text | Y=1) . P(Y=1)

↓

P(Y=1) . $\prod_{i=1}^{d}$ P(w_i | Y=1)

P(Y=0 | text)

↓

P(text | Y=0) . P(Y=0)

P(Y=0) . $\prod_{i=1}^{d}$ P(w_i | Y=0)

no

no

compare

higher no → predict

Real world

authorized

Hi, how are you

algo → very simple

→ making assumptions ⇒

Training time

Lin Reg^n → params Training

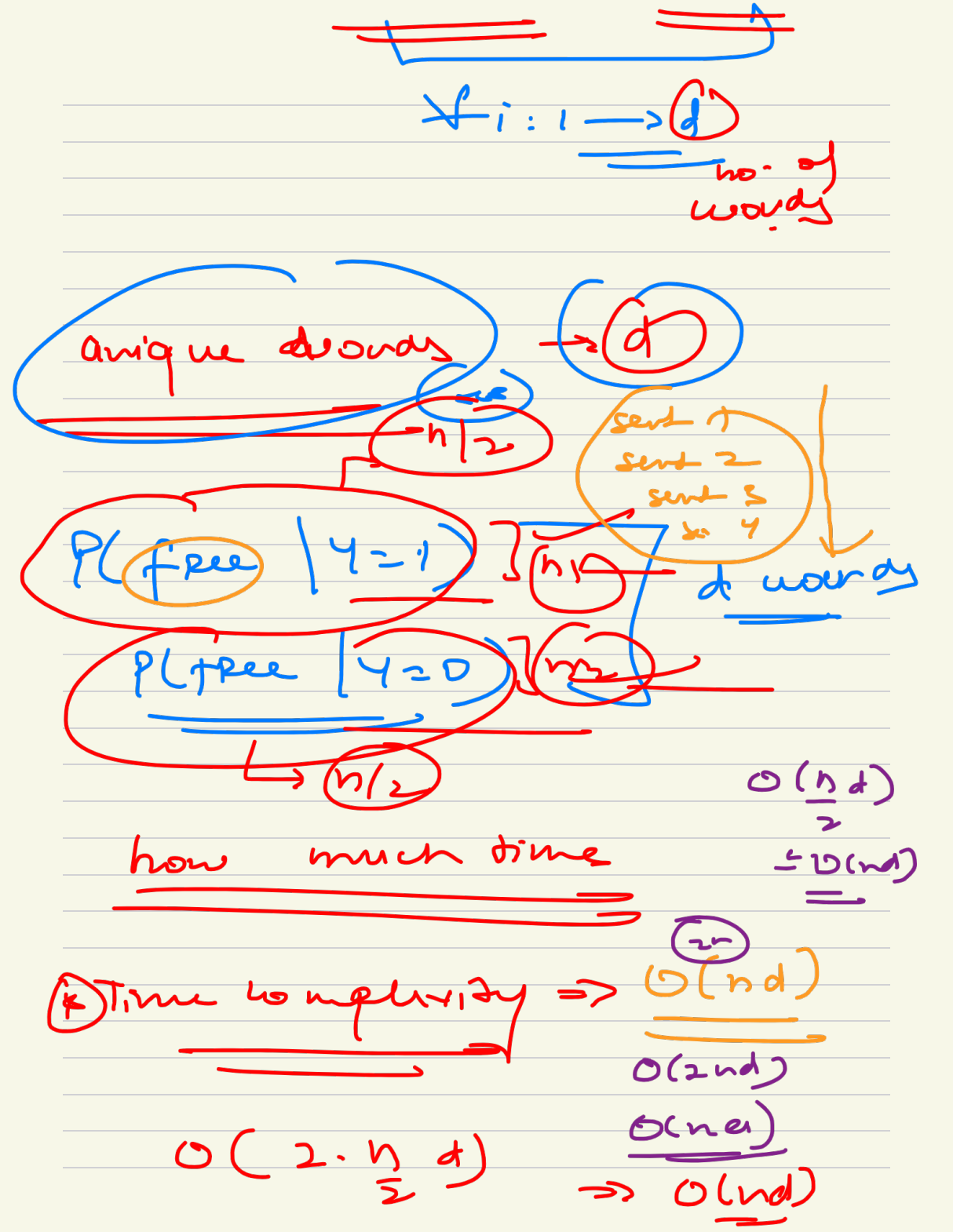
NB → prob.

Pre compute probabilities

→ prior: $P(Y=1)$, $P(Y=0)$

→ likelihood prob.

⇒ $P(w_i|Y=0)$, $P(w_i|Y=1)$


∀ i : 1 → d
no. of words
unique words
d
n/2
sent 1
sent 2
sent 3
sent 4
d words
P(free | y=1)
n/2
P(free | y=0)
n/2
n/2
how much time
O(nd)
2
= O(nd)
* Time complexity =>
O(nd)
O(2nd)
O(2 · n/2 d)
O(nd)
=> O(nd)

$O(n \cdot d)$

Classes $\Rightarrow$ $O(n \cdot d \cdot c)$

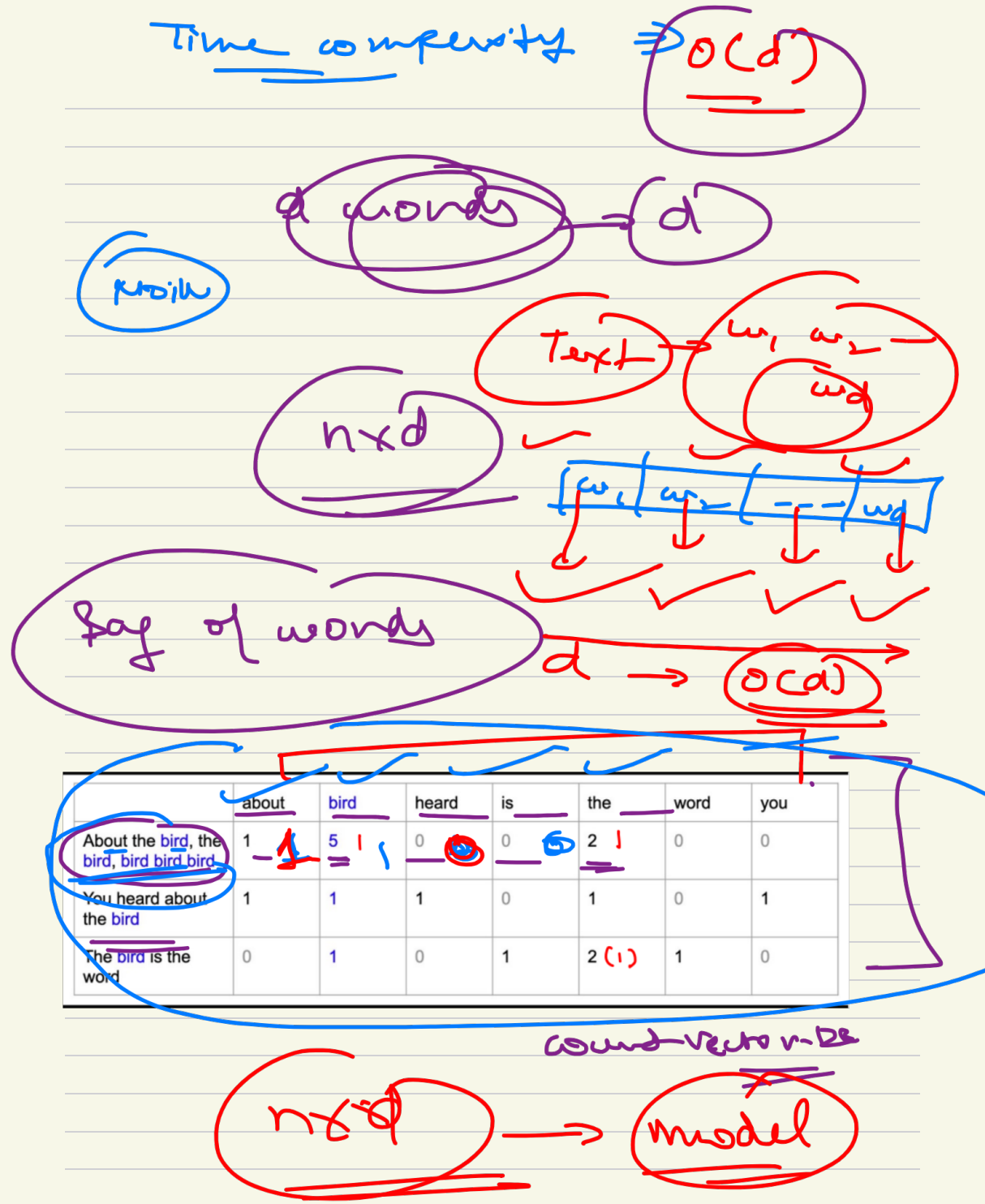
$\Rightarrow O(n \cdot d)$ # classes

2 $\rightarrow$ +ve, -ve

NoW $\rightarrow$ # unique words = # of features

Test Phase $\Rightarrow$ PRE on

$$P(y=1 \mid \text{text}) = P(y=1) \prod_{i=1}^{d} P(w_i \mid y=1)$$


Time complexity
O(d)
d words
d
Text
nxd
Bag of words
d → O(d)
about bird heard is the word you
About the bird, the bird, bird bird bird
You heard about the bird
The bird is the word
Count-Vectorizer
nxd
model

Bag of words

